Password Manager with Multi-Factor Authentication

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Abstract— Data breach is a serious issue as it leaks the personal information of more than billions of users and their privacy is compromised. More than 77% of organizations do not have a Cyber Security Incident Response plan. So, it is necessary to be informed of network security and ways to store and generate secured passwords. Having different and random passwords for ones digital accounts can exponentially increase the security of user's data. The goal of this project is to build a password manager which can securely store and encrypt passwords and other data. The multifactor authentication will provide increased security to validate the user into password management system. Multifactor authentication system includes physical security key and graphical password authentication.

Keywords: Cyber Security, Cryptography, Multi-factor authentication, Master Password.

1.INTRODUCTION

Passwords act as the foremost security for one's digital information against unauthorized access. When the password is stronger and secured, it becomes a herculean task for malicious software and hackers to brute force attack. It is mandatory to have strong passwords for all digital accounts [5, 6].

Characteristics of a strong password: It should be more than eight characters. One should not use personal information. Use unique and random passwords. Avoid consecutive keyboard combinations. Include special characters and numbers. Use mixture of uppercase and lowercase letters. Change passwords regularly.

When the user signs up with various digital systems using several different passwords, it becomes very hard to remember them. To overcome this difficulty password manager is used. With password manager one no need to remember all the passwords. The user can generate and store random passwords. [8]
This doesn't mean the password manager is flawless. There are two most common ways to store the password. The first way is to store the passwords in the internal storage of the application. Here the password is not synchronized. So, if the user changes the device, one should manually port the passwords to the new device. All passwords stored on the device are permanently inaccessible and cannot be recovered in the event of theft or loss. The second way is to store the password in an online database like the cloud. Here the passwords are perfectly synchronized and available on multiple devices simultaneously. But the credentials for every device will be exposed if a data breach happens in the online database. With the leaked password, cyber criminals can hack into the users account and steal ones sensitive information, bank account details and digital identity. Having multi-factor authentication to login in, to password manager will provide additional security and makes to difficult for hackers to access the sensitive information.

2. LITERATURE SURVEY

The users need to enter three criteria at first like a phrase, clue, and username. In this, two of the criteria remain the same and the user can change any one of them. From this method, the user needs to remember only two of the criteria, instead of all three of them. Creating distinct passwords completely based on the clue. Whenever the user wants to access the information in the application where one previously used the password by this approach, one should enter the clue one used earlier to identify that application and the system will recreate the password for them. [9] By this approach even if the user selects the exact password once again, it will be different from the previously hashed one. So, it can weaken hash table attacks. For this reason, salt can be used in hashing instead of these different criteria and the user no need to remember additional information other than the master password. [1]

A password manager that ensures the merging of key values of password managers and provides graphical passwords to make a solution for the existing problems of storing passwords and relating passwords with accounts. It uses image cues to remember graphical passwords Instead of remembering actual passwords one by one. These cues help users to remember the passwords in an easier way and to link passwords with different accounts in a better way. By

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enabling the user to use the same visual cue for several accounts, it also offers password reuse in a secure manner. The password is scrambled differently for each account when a user appends the relevant password cue to several accounts, which encourages secure reuse. It gives better memory, while better managing user's passwords and the link between passwords and accounts. It also improves security by using user habits and they also assist users with current habits like password reuse. [14]

According to several high-level password management results, it is not necessary to store passwords with a high level of entropy in order to retrieve them [4], [10], and [15]. By hashing the (password, domain) pair and registering it as a strong password with the server, PwdHash [10], for instance, converts low entropy-master passwords to high entropy-store passwords and retrieves passwords. PwdHash converts a user's real password deterministically into a more challenging password, however this change does not provide protection against offline dictionary attacks. However, if any user makes use of easy to recollect password with PwdHash for numerous digital accounts, the data breach of even one web server succeeds in the leak of a password by a dictionary attack and then calculates users all actual passwords which are derived from a password[2, 6, 11, 13, 18].

An MBCCO scheme password manager, the use of a mobile device as a security token and an additional device for decryption is one of the essential aspects of this strategy that makes it vital to note the difficulty in extracting the encrypted data and the key used for decryption. Thus, a secure secret key in the system encrypts the registered login information before sending and storing the encrypted data on the connected device to the PC and erasing it [17, 19, 20]. That is, other from the login credentials, the PC is where the single secret key is kept. The PC does not save the login details. The user chooses the web service from the register on the device each time they attempt to log into a registered web service. The user-selected encrypted login information is then transmitted to the PC, where it is further decrypted using a secret key. The user can then log in the web service. Has an additional reliability security token is used here.[1,2,3]

3.ARCHITECTURE

Architecture of the system mainly consist of 4 modules A) Master password B)Vault Key C) Multi-Factor Authentication D) Web server.

The process of confirming a user's or a digital system's integrity is known as authentication.

It validates the entities as to whom or what they claim to be.

A) Master password: Having access to additional passwords requires a master password. With the use of a master password, the user may access the encrypted file containing all of their passwords. The master password is hashed to save it securely. With hashing, one can convert any size of data into a fixed-length. The purpose of hashing is once the master password is hashed it cannot be reversed, unlike encryption.
PBKDF2 hashing program is used in password manager.

Password-Based Key Derivation Function 2 (PBKDF2) is a key derivation function that uses padding and numerous iterations of the HMAC algorithm.

Reason to use PBKDF2 hashing:
· Both dictionary and rainbow table assaults cannot harm it.
· Since it is slower by design, it is more difficult for hackers to guess the master passwords.
PBKDF2 takes the below **input arguments** and displays the resultant output as derived **key.**

B) pbkdf2 is the vault key (password, salt, iterations-count, hash-function, derived-key-len)

This vault key is used in **AES-256** to encrypt and decrypt other passwords. AES is a symmetric key cipher. It consists of a sequence of combined operations, some of which involve rearranging bits throughout (permutations) and displacing inputs with specific outputs (substitutions).

AES-256 is a 256-bit key that is used to encipher and decipher messages. It is widely known as military grade encryption.

Its performance is optimal and faster when compared to many other cryptographic algorithms. It is resistant to most of the cryptographic attacks.

C) Multi-Factor Authentication
It is a verification protocol that requests two or more verification parameters from the user in order to provide access to a system.

Biometric authentication depends on the distinct physical nature of a user for verification. The system will use biometric services like a fingerprint scanner, facial recognition, or retinal recognition to authenticate the user along with the master password.

Hardware-based tokens like physical security keys can also be used to authenticate the system.

D) Web server:
The vault key is appended and master password and then hash it with the PBKDF2 algorithm to authenticate the user in the web server. Along with this, multi-factor verification is used for additional security.

User Authentication = pbkdf2 (vault key | master password)

Cloud Firestore is used to store the encrypted password and it can only be decrypted after the user authentication process.

Cloud services for the database functionality which is needed to be accessed from anywhere without any disruption and the

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password will be synchronised across all devices of the user that contains this system.

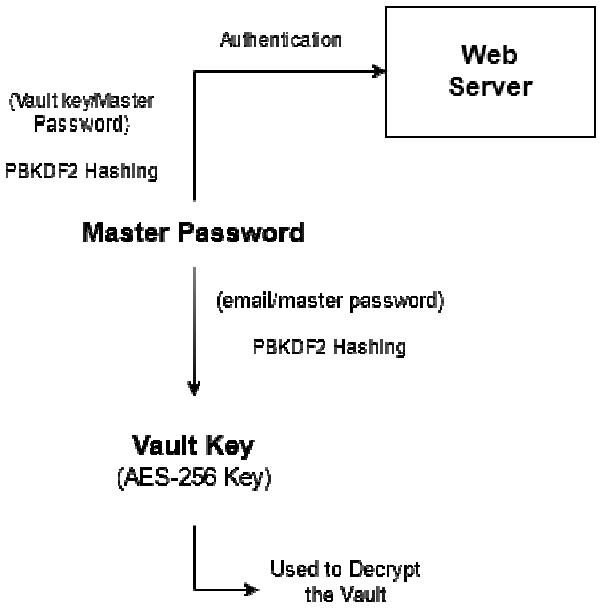


Fig 3 – Module diagram

4.IMPLEMENTATION

4.1 Proposed password manager types:

One of the aspects of the application is it provides the users the option to choose what type of password manager, they want it to be.

That is the user will be provided 2 choices:

a. Offline Password Manager

b. Cloud-based Password Manager

a. When the user chooses offline password manager, all their passwords will be encrypted using AES-256 encryption and it will be stored locally in the device and it can be decrypted only by providing the master password. It cannot be accessed elsewhere.

b. When the user chooses a Cloud-based password manager, all their passwords will be encrypted and store in the cloud. Here all the encrypted passwords are stored in the cloud in real-time, and they can be accessed across multiple devices using this application.

The user can select the password manager type based on their convenience. If they trust to store their password only locally, the offline password manager will be the suitable option. However, the passwords cannot be recovered if the device is lost or stolen.

On the contrary, to common belief, it is safer to store the password in the cloud service and one can't access the data, since all information is encrypted before storing in cloud database. Only the user can decrypt it with the master password.

Additionally, the hashed master password will only be kept on the cloud, and when a user signs in, the hashed master password is compared to verify their identity. The systems don't have access to master password as plaintext either. In the worst case, if the cloud database is breached, the hacker will have only the hashed master password which cannot be converted back to plaintext and without the master password, the user's other passwords cannot be decrypted. Hence, the privacy and personal information of the users will remain safe and it cannot be accessed.

4.2 New approach to store master password:

A new method to store master password with an example and block diagram

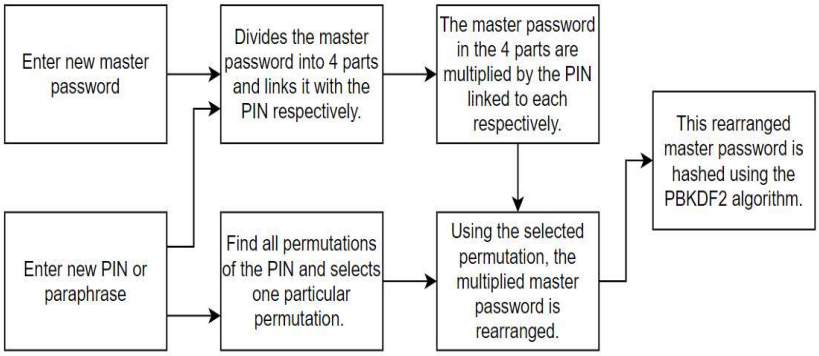


Fig. 4.2 – Block diagram for storing master password

a. Let the master password be simple as "snow" for easy understanding.

(Note: The real master password of the user should be distinct to prevent brute-force attack)

b. Let the PIN be: 1 2 3 4. Now the master password "snow" is divided into 4 parts and linked with the PIN respectively.

c. For the understanding purpose it can be represented as ["1:s","2:n","3:o","4:w"]

d. Find all the permutations without repetition of the PIN 1 2 3 4. Out of 24 possible permutations, one permutation is selected based on a condition.

e. Let the selected permutation be: 4 3 2 1

f. The master password in the 4 divided parts is multiplied by no. of times of their respective PIN.

g. Now, this can be represented as ["1:s","2:nn","3:ooo","4:wwww"]

h. With the order of the selected permutation, the above pair can be rearranged as follows:

["4: wwww", 3: ooo"," 2: nn"," 1: s"]

i. Now, in rearranged pair, the master password is concatenated as: "wwwwooonns"

j. This rearranged master password "wwwwooonns" is hashed using the PBKDF2 hashing algorithm.

Once, this rearranged master password is hashed it cannot be reversed and only the hashed master password is stored. As a result, only the person who is familiar with both the master password and PIN may access the system and read other passwords. The PIN will not be stored anywhere.

4.3 Interface of password manager:

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4.3.1 For signing up:

Initially, the users should sign-in with the username and create a new master password. Certain parameters is set, like it should contain at least one capital letter, symbols, numbers and it should be of a certain length to make sure the master password is strong. The user must keep in mind the master password. This master password is hashed using the PBKDF2 algorithm.

Then the user will have to enter a paraphrase with 4 words or a pin code and remember it. This will be used as multi-factor authentication and to validate the master password while logging in. The above details will be enciphered using the AES-256 hashing algorithm and stored locally in the device and also in the cloud if the user desires.

Once, the user has successfully signed in to the application, they can choose the type of data to store. For example, one can store the username and password of an email account. The credit card and debit card information, secure notes can also be stored. These pieces of information will be encrypted and cannot be viewed without logging in.

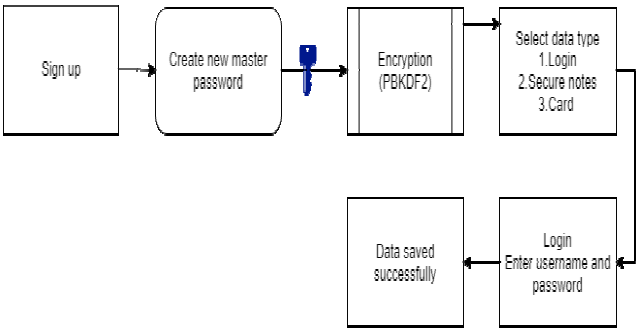


Fig. 4.3.1 – Sign up

The user can also use one's fingerprint while authenticating the application instead of the master password. This feature can be enabled while signing up to the application or later in the setting if the device contains a fingerprint scanner.

4.3.2 The functions of the password manager:

After logging in, the user can select the types of information they want to store. There will be a menu where you can see all the data and passwords that have been saved. The password is decrypted using the AES-256 technique and presented, for instance, if the user chooses to view the account's user name and password. The secure notes and card information can be viewed using this same procedure.

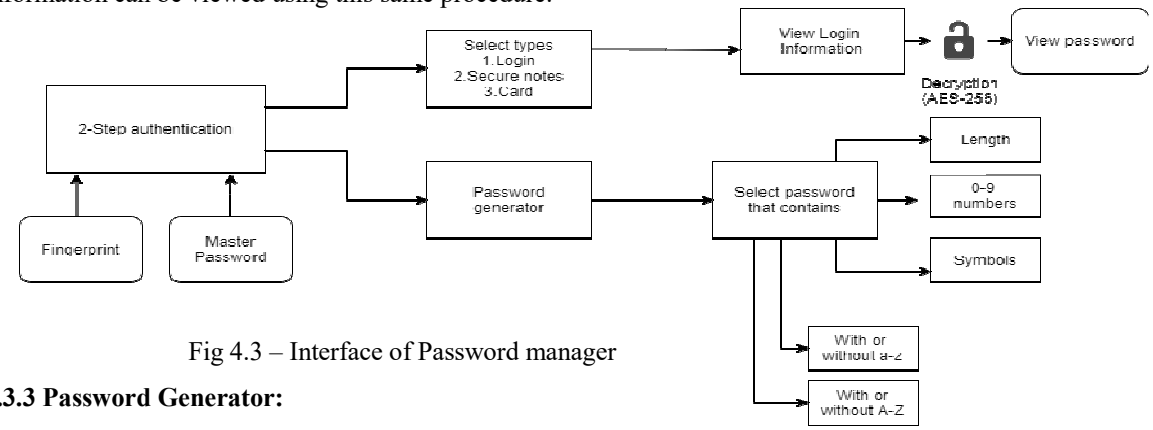


Fig 4.3 – Interface of Password manager

4.3.3 Password Generator:

To help users build random, one-of-a-kind passwords, the programme will have a password generator. This application can generate random paraphrases besides random passwords. These paraphrases and passwords are strong enough to mitigate brute force attacks. One of the concerns of using paraphrase is it is susceptible to dictionary attack.

The paraphrase this application generates contains 4 words that are not commonly used in vocabulary. This method mitigates the dictionary attack as most of the attack is performed by brute-forcing the top words available in the dictionary. The users can also mix other languages that use the Latin alphabet like French, Italian, and Spanish to generate the paraphrase if they want additional security. In this way, it is nearly impossible to brute attack or dictionary attack the generated paraphrase.

On the other hand, the user can impose a variety of restrictions to produce a robust password. The password's length can be customised by the user. The user can choose whether or not a password needs symbols, special characters, or numbers. It is also possible to specify the length of these characters. It is also possible to state for the user to generate a password with capital letters.

Since the password manager generates unique and random passwords it is not in any way related to the user's personal information or something publicly available. This makes the generated password resistant to social engineering attacks. That is, it is not possible to psychologically manipulate the users to reveal their personal information related to their password, unknowingly.

4.3.4 Settings in password manager:

The user can set the vault timeout period by which the application locks itself exceeding the time. One should again log in to the system to regain access to the information. This increases the security of the system.The settings to enable or disable fingerprint authentication or PIN code are available here. The user can also manually sync their encrypted information to the cloud database.

The multiple-step log-in service like authenticator app, hard security tokens, and physical security key can also be enabled here. The user can also change the master password or paraphrase here and a new vault key will be automatically generated.

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5. Future Scope

The application design should be user-friendly and simple to use. This system will be available on all major platforms. Since all the passwords are available in a single system it is at most important to penetration test it with different techniques. Additional hashing techniques like Scrypt will be added. The application will be open-source. The security analysts and coders can test the system and find vulnerabilities. By fixing these security bugs, the security of the methodology can be increased.

6. Conclusion

The password manager is the necessary application to protect one's privacy and secure passwords. This approach is suggested to strengthen authentication security in password managers. The drawbacks of this authentication system are there is a possibility of many security bugs as the system is new. In further improvements these problems will be solved.

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